PHYC10004 Physics 2: Physical Science & Technology Exam 2017 Solutions

Question 1

a) Forces equal in magnitude and opposite direction diagram

b) $F_{13} = F_{23}$

$\frac{kq_1q_3}{(2d)^2} = \frac{kq_2q_3}{d^2}$

$\frac{q_1}{4} = q_2$

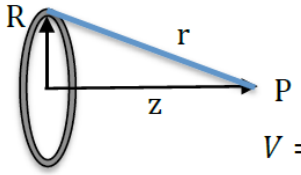
Therefore $q_1 = 4q_2$

c) $V = V_1 + V_2 \ldots$

$V = \frac{kq_1}{d} + \frac{kq_3}{d}$

$V = -\frac{9\times10^9\times10\times10^{-9}}{0.05} + \frac{9\times10^9\times2\times10^{-9}}{0.05} = -1440V$

Question 2



a) $\Delta V = \frac{1}{4\pi\epsilon_0}\frac{\Delta Q}{r}$

$V = \sum \Delta V = \frac{1}{4\pi\epsilon_0}\sum\frac{\Delta Q}{\sqrt{z^2+R^2}} = \frac{1}{4\pi\epsilon_0}\frac{Q}{\sqrt{z^2+R^2}}$

$r = \sqrt{z^2+R^2}$

b) $E = -\frac{dV}{dz} = -\frac{d}{dz}\frac{1}{4\pi\epsilon_0}\frac{Q}{\sqrt{z^2+R^2}} = \frac{Q}{4\pi\epsilon_0}\frac{z}{(z^2+R^2)^{3/2}}$

Question 3

a) same symmetry as wire

at ends E is perpendicular to surface vector => $\int E \cdot dA = 0$

curved surface, E is parallel to surface vector => $\int E \cdot dA = E\times A$

This is much more complicated for rectangular prism

b) $\oint E \cdot dA = \frac{q}{\epsilon_0}$

E.dA=0 at ends

$E\int_{curved\ surface} dA = \frac{q}{\epsilon_0}$

$E2\pi rL = \frac{q}{\epsilon_0}$

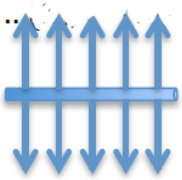
$E = \frac{q}{2\pi rL\epsilon_0}, \lambda = \frac{q}{L}$

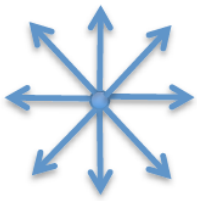
$E = \frac{\lambda}{2\pi\epsilon_0 r}$

c) $E = \frac{20\times10^{-9}}{2\pi\times8.85\times10^{-12}\times0.08} = 4500\ V/m$

d)

Non-uniform ... each diagram





Question 4

a) $C = \frac{2U}{V^2} = \frac{2\times400}{2000^2} = 2\times10^{-4}F$

b) $C_0 = \frac{A\epsilon_0}{d} = \frac{5\times8.85\times10^{-12}}{5\times10^{-5}} = 8.85\times10^{-7}F$

$\kappa = \frac{C}{C_0} = \frac{200\times10^{-6}}{8.85\text{x}10^{-7}} = 225$

c) No -

$E = \frac{V}{d} = \frac{2000}{5\times10^{-5}} = 4\times10^{7}V/m$

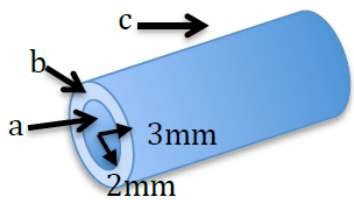
Dielectric strength < E_{max} hence
Breakdown of capacitor will occur

Question 5

a) Junction: $I_1 + 2 = I_2$
Loop 1: $10 - 8I_1 + 2\times3 = 0$
Loop 2: $\varepsilon - 2I_2 - 3\times2 = 0$
$I_1 = 2A$.
$\varepsilon = 14\,V$

b) No while the globe's resistance in this configuration can be determine by V=iR, for different voltages or currents, R may change , Ohm's Law defines an ohmic device to have a constant R, more tests would need to be done to confirm this.

Question 6



a) hollow wire
inner radius = 2mm
outer radius = 3mm

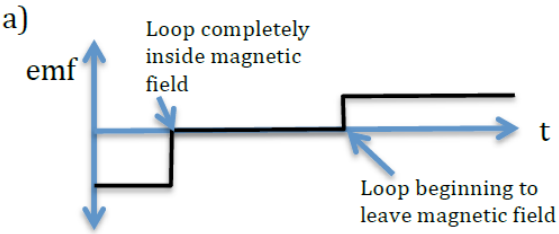
Requires the correct placement of a, b, c

b) at surface of wire r=3mm and B=33μT
Ampere's Law $\oint B \cdot ds = \mu_0 I$
$B2\pi r = \mu_0 I$
$I = \frac{B2\pi r}{\mu_0} = \frac{33\times10^{-6}\times2\pi\times3\times10^{-3}}{4\pi\times10^{-7}} = 0.495A$

c) extra term - μi_D for the displacement current This current comes about due to the change in electric flux causing a changing B field . This can be demonstrated in capacitors or electromagnetic fields – light

Question 7

a)



b) C

c) loop area = $(0.1)^2 = 0.01 m^2$

$$\varepsilon = -\frac{d\phi}{dt} = -A\frac{dB}{dt} = -A\frac{d}{dt}(0.7 - 0.4t^2) = A(2\times 0.4t) = 0.01\times 0.8\times 2$$
$$= 0.016\ V$$

d) $\varepsilon = BA\omega\sin\omega t = 0.7\frac{\Delta A}{\Delta t}$, $\Delta A = 0.01 - 0 = 0.01$

$\varepsilon_{max} = BA\omega = BA2\pi f$

$f = \frac{\varepsilon_{max}}{2\pi BA} = \frac{3}{2\times\pi\times 0.7\times 0.01} = 68.2\ Hz\ or\ turns/second$

e) increase number of loops or increase area of loop

Question 8

a)

F = weight – buoyancy

= mg – ρVg

= 686 N – 303 N

= 383 N

b)

F = gravitational force

= mg = 686 N

c)

Mass of displaced water = difference of masses

= 14.70 kg – 13.74 kg

= 0.96 kg

Volume of water or shield = mass/density

= 0.96 kg / 1030 kg m^3

= 0.93 * 10^-3 m^3

Density of shield = mass/volume

= 14.70 kg / .00093 m^3

= 15.8 * 10^3 kg/m^3

=> Shield some combination of lead and gold

d)

Bernoulli equation: P1 + 1/2 ρ v1^2 + ρgh1 = P2 + ½ ρ v2^2+ ρgh2

200 * 1000 Pa + ½ * 1000 kg/m^3 * (1 m/s)^2 = 101.3 * 1000 Pa + ½ * 1000 kg/m^3 * v^2

V = 14.1 m/s

Question 9

a)

Ice: mcT + mL = (0.4 x 2220 x 2K) + (0.4 x 333000) J

= 1776 J + 133200 J

Water: mcT = (1 x 4186 x 21K) = 87906 J

➔ Not enough E to melt all ice, so end result have both:

87906 J – 1776 J = 86130 J = Energy remaining to melt ice at T=273K
86130 J = m x 333000 → m = 0.259 kg ice melted to water
Therefore, final result will be: 141g of ice and 259g of water at T = 0°C

b)
C = Change in energy / (change in temp * mass)
= 487.5 J / (50 degrees * 25 g)
= 0.39 J/g deg C

c)
i)
Energy = work + thermal
= 0 + 1500 = 1500 J
ii)
Energy = work + thermal
= 1500 + 0 = 1500 J

d)
i)
Ideal gas equation: PV = nRT
V = 62.5 * 8.31 * 300 / 10^5 = 1.56 L
ii)
Ideal gas equation: T1/P1 = T2/P2
P2 = 263/300 * 10^5 = 8.77 * 10^5 Pa

Question 10

(a) (mislabeled c) (i) Photoelectric effect finding the frequency:
Workfunction Φ of Na = 2.28 eV → Φ = hf → f (Na) = 5.51e14 Hz
Workfunction Φ of Cs = 2.14 eV → Φ = hf → f (Cs) = 5.17e14 Hz

(a)(mislabeled c) (ii) Knowing E = hf – Φ, smaller Φ leads to higher KE of electrons → Caesium will have higher maximum KE.

(b) If diffraction to occur, Ferrari must require its *de Broglie* wavelength to be of a similar order of magnitude as the underpass gap, L = 10 m.
That is:

$$\lambda_{dB} = \frac{h}{p} = \frac{h}{mv} = L$$

which gives $v = \frac{h}{mL} = 5.53 \times 10^{-38}$ m/s.

(c) $\Delta p / p = 10^{-6}$ implies $\Delta p = 10^{-6}\, p$
Uncertainty principle: $\Delta x \, \Delta p = h / 4\pi$ implies $\Delta x = \frac{h}{4\pi \, \Delta p}$
and we have $p = \frac{h}{\lambda}$ for the photon, so

$$\Delta x = \frac{h}{4\pi(10^{-6}\, p)} = \frac{\lambda}{4\pi 10^{-6}}$$

With λ = 600 nm, we get Δx = 0.047 m.

Question 11

(a) (i) Using Wien's Law $\lambda_{MAX}T = 2898 \times 10^{-6}$
$T = 2898 \times 10^{-6} \div 500 \times 10^{-9}$
$=5796$ K (or 5800 K)

(a) (ii) Use Stefan's Law $P = \sigma\varepsilon AT^4$
$= 5.67 \times 10^{-8} \times 1 \times 6.09 \times 10^{18} \times 5796^4$
$= 3.90 \times 10^{26}$ W

(b) (i) Use $E = hc/\lambda$
$= 6.63 \times 10^{-34} \times 3.00 \times 10^8 \div 500 \times 10^{-9}$
$= 3.98 \times 10^{-19}$ J *or* 2.48 eV if other value for Plank's constant is used

(b) (ii)
use n = power divided by energy of one photon
$= 3.90 \times 10^{26} \div 3.998 \times 10^{-19}$
NOTE: must use joule value of photon energy here
$= 9.80 \times 10^{44}$ photons/sec (no unit required)

Question 12

(a) (i) Use $hf_{LIMIT} = hcR(1/2^2)$ (having set n_1 to ∞)
$= 6.63 \times 10^{-34} \times 3.00 \times 10^8 \times 1.097 \times 10^7 \div 4$
$= 5.45 \times 10^{-19}$ J *or* 3.40 eV

(a) (ii) Use $r_n = a_0 n^2$
$= 52.92 \times 10^{-12} \times 10^2$
$= 5.29$ nm or 5.29×10^{-12} m

(a) (iii) Use $v_n = c\alpha_{EM}/n$; α_{EM} the fine structure constant
$= 3.00 \times 10^8/(137 \times 10)$
$= 2.19 \times 10^5$ m s^{-1}
NOTE: students may alternatively use 0.007297 for the fine structure constant

(b) Emission spectra are generally observed from hot gases where the higher energy levels are populated. Absorption spectra are generally observed when the photons pass through a cool gas and excite transitions from low n values to higher n values. However in cool gases very few atoms are likely to be in any levels other than $n = 1$ so the transitions *to* levels with $n > 1$ are highly unlikely to be produced. (something along these lines)

- needs to be reference to both emission and absorption spectra

(c) (i) Use $E = -13.6Z^2\left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right)$ with $Z = 23$; $n_1 = 1$ and $n_2 = 2$
$= 13.6 \times 23^2/4$ (or the Rydberg formula)

= 1799 eV or 1.80 keV or 2.88×10^{-16} J

(c) (ii) 1799 ev – ionization energy of n = 4 level = KE; answer in J also acceptable

Calculation of ionization energy of n = 4 level →

$E = -13.6Z^2\left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right)$ with Z = 5, n_2 = 4 and $n_1 = \infty$);

E = 85 eV (or value in J)

Final difference in E = 1714 eV or 2.74×10^{-16} J

Question 13

(a) 13 g of pure N-13 will have 6.02×10^{23} atoms (Avogadro's number); hence number = $(1.49 \times 10^{-6}/13) \times 6.02 \times 10^{23} = 6.90 \times 10^{16}$. (Show reasoning *clearly*!)

(b) Initial activity = λN_0; hence need value for λ.

$\lambda = \ln 2/\tau_{0.5} = 0.00116$

$= 0.00116 \times 6.90 \times 10^{16} = 8.00 \times 10^{13}$ Bq

(c) Use $N(t) = N_0 e^{-/t}$

$= 6.90 \times 10^{16} \times \exp(-0.00116 \times 3600)$

$= 1.06 \times 10^{15}$ atoms

(d) Use $r = (1.2 \times 10^{-15}) \times A^{0.33}$

$= 1.2 \times 10^{-15} \times 13^{0.33}$

$= 2.82 \times 10^{-15}$ m

(e) Options (some non-physical, but not established with students)

- ${}^{13}_{8}O \longrightarrow {}^{13}_{7}N + {}^{0}_{1}b + n$ positron emission
- ${}^{13}_{8}O + {}^{0}_{-1}e \longrightarrow {}^{13}_{7}N + n$ electron capture
- ${}^{13}_{6}C \longrightarrow {}^{13}_{7}N + {}^{0}_{-1}b + \bar{n}$ negative beta decay

The first two are what actually happens; the third does not happen because C-13 is stable. Correct neutrino required.

(f) One possibility (explanation required may have other options possible):

The binding energy is the amount of energy required to 'unbind' a nucleus into its individual nucleons. The larger the binding energy is, the harder it is to 'unbind' it; hence the higher binding energy the more stable the nucleus is, and vice versa.